AP018223.1:1024859-1037851
Nostoc linckia NIES-25

Acetylase Orn_Arg_deC_N DAC Prenyltransf ParD_antitoxin DUF3143 DnaJ ParE_toxin RelB-like DnaJ DnaJ_C HSP70 Peptidase_M28 Peptidase_M28 ? ? DUF559

CADCTY010000587.1:899-893

uncultured Leptolyngbya sp.

? ParE_toxin

CDRZ01000260.1:89-5829
Syntrophaceticus schinkii

? DeaD_dimer ParE_toxin Mg_chelatase_C DUF6075 PemK_toxin DUF6364 ? HigB-like_toxin HTH_3 Peptidase_M78

BEHR01000009.1:71700-83957
bacterium HR15

? Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C Clp_N Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small ? ? ? RelB-like ParE_toxin ? ? ? ? ?

DNKF01000157.1:4217-4045
Methylococcaceae bacterium

META META ParE_toxin DUF6290 Aminotran_1_2 Aminotran_1_2 UvrB_inter ResIII

DOWK01000431.1:147-7027
Blastocatellia bacterium

Acetyltransf_11 Hexapep Hexapep Hexapep Hexapep ParE_toxin RelB-like FAD_binding_4 FAD-oxidase_C PIN PhdYeFM_antitox CHRD FG-GAP_3 FG-GAP_3 FG-GAP_3 SDF Uma2

DQWU01000172.1:2537-3341
Candidatus Bathyarchaeota archaeon

AAA DSHCT ParE_toxin

DQZL01000585.1:252-2238
Thioploca sp.

ANG-1 CTHRC1_C ParE_toxin DUF2281

DRAW01000122.1:150-2716
Methanomicrobia archaeon

DSHCT ParE_toxin ? Aminotran_5 NifU_N AIRC

DASYYM010000973.1:236-9771
Puia sp.

? ORF6N Phage_integrase Phage_int_SAM_1 Phage_integrase Mg_chelatase Toprim_2 DNAG_N Zn_ribbon_DnaG HTH_3 RelB-like ParE_toxin ? ?

DASZBR010000005.1:86063-98211
Gammaproteobacteria bacterium

TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C YecM Pyr_redox_2 HTH_1 LysR_substrate HTH_18 GyrI-like Beta-lactamase RelB-like ParE_toxin ? MFS_1 Dabb HTH_1 LysR_substrate Acetylase Acetylase Fic DUF4172

DASZFI010000197.1:3741-6146
Gemmataceae bacterium

Proton_antipo_N Oxidored_q2 Oxidored_q3 NADHdh SMC_ScpB ParE_toxin PhdYeFM_antitox NTP_transferase LbH_GLGC

DQRP01000035.1:118-1829
Aquifex sp.

URO-D Unstab_antitox ParE_toxin PGM_PMM_I

DQTY01000374.1:94-1676
Anaerolineae bacterium

Methyltransf_11 ParE_toxin PhdYeFM_antitox PhdYeFM_antitox

DYDP01000109.1:150-1516
Rickettsia endosymbiont of Bembidion lapponicum

ParE_toxin PhdYeFM_antitox ?

DATEXL010000273.1:1706-8609
Polyangia bacterium

PhdYeFM_antitox ParE_toxin Kelch_4 Kelch_KLHDC2_KLHL20_DRC7 Kelch_4 Lipocalin_2 Methyltransf_23 Glyco_trans_1_2 GH97_N Glyco_hydro_97 GH97_C

DATFWS010000259.1:560-4222
Candidatus Binatia bacterium

? Rap1a SWIM PhdYeFM_antitox ParE_toxin Peptidase_S74 ABC_tran_CTD ?

DATDOF010000804.1:310-2816
Chitinophagaceae bacterium

HTH_18 AraC_binding ParE_toxin RelB-like HTH_18

DATGLZ010000295.1:1-5113
Chloroflexota bacterium

BPD_transp_1 BPD_transp_1 PhdYeFM_antitox ParE_toxin Sulfatase Glyco_hydro_39 LacI Peripla_BP_3

DATHAT010000173.1:1-3147
Tepidisphaeraceae bacterium

HATPase_c_3 ? ParE_toxin ? ? Peptidase_M29

DATGVO010000021.1:1-3797
Planctomycetaceae bacterium

Receiver Receiver His-Kinase His-Kinase PAS PAS Receiver ? PhdYeFM_antitox ParE_toxin

DATGUL010000083.1:181-9849

Sphingobacterium sp.

Rep3_N Rep3_C ? CbiA AbiEii ? ? PhdYeFM_antitox ParE_toxin ? ? HTH_17 PIN AbiEii AbiEii AbiEi_3_N

DATGSN010000520.1:1-1774

Jiangellaceae bacterium

? ? PhdYeFM_antitox ParE_toxin

DATGFC010000106.1:49934-62951
Candidatus Sulfotelmatobacter sp.

HIT S1 S1 S1 S1 S1 S1 Pyr_redox_dim Pyr_redox_2 HicA_toxin HicB_lk_antitox LplA-B_cat E1_dh ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr ? Cytochrome_CBB3 Amidohydro_3 HTH_31 Cupin_2 Fasciclin

DAOLZS010000428.1:120-2929
Anaerolineales bacterium

dNK ? RelB-like ParE_toxin HD

DATIGG010000362.1:400-4082
Thermodesulfobacteriota bacterium

HicA_toxin HicB_lk_antitox ParE_toxin PhdYeFM_antitox PhdYeFM_antitox HTH_3 PIN RHH_1 PIN RHH_1

DATKAE010000023.1:3560-5008
Thermoanaerobaculia bacterium

STY-kinase PD40 PhdYeFM_antitox ParE_toxin TPRs

DATJXG010000114.1:1-6339

Terriglobales bacterium

Calcineurin ParE_toxin PhdYeFM_antitox Transketolase_C Transket_pyr Cytochrome_CBB3 Trans_reg_C Receiver His-Kinase His-Kinase HAMP ?

DATLTK010000869.1:1-2775
Chloroflexota bacterium

? ParE_toxin RelB-like Gp53-like_C SH3_3

DATKJU010000297.1:106-4535
Thermodesulfobacteriota bacterium

Fer2_3 Fer2_2 ParE_toxin PhdYeFM_antitox Guanylate_cyc TPRs TPRs LemA

DASZYQ010000048.1:39896-56183
Mucilaginibacter sp.

Endonuc-EcoRV ? ? Peptidase_S13 AA_permease_2 H_PPase DUF5686 CarbopepD_reg_2 ParE_toxin RelB-like AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C2 GFO_IDH_MocA_C2 DHDPS Hc1 Glyco_hydro_125 Aminotran_1_2

DASZRN010000071.1:6431-20347

Polyangia bacterium

Sigma70_r2 Sigma70_r1_2 ? ? CN_hydrolase LNT_N PNPOx_N UPF0020 THUMP RlmL_1st Unstab_antitox ParE_toxin ATP-synt_DE_N ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt ATP-synt_ab_C ATP-synt_ab ATP-synt_ab_N OSCP ATP-synt_B ATP-synt_B

DASZPW010000203.1:299-3527

Chitinophagaceae bacterium

RelB-like ParE_toxin Peptidase_S37 Pyr_redox_2

DATMDN010001211.1:89-2627

Polyangiaceae bacterium

ParE_toxin PhdYeFM_antitox Sigma70_r2 Sigma70_r4_2 ? ?

DATNEW010000020.1:53208-66841
Gemmatimonadaceae bacterium

? HhH-GPD DUF6496 PAP2 BTAD GerE ? ParE_toxin PhdYeFM_antitox ? ? MutS_V PAP2 FUSC-like FUSC_2 HlyD_D23

DASRKP010000486.1:92-9200
Candidatus Limnocylindrales bacterium

OTCace_N OTCace Pribosyltran DUF4032 DUF5615 DUF433 N_methyl SBP_bac_10 ParE_toxin RelB-like FeS_assembly_P ParA Glyco_hydro_43 BNR_2 ? ? ?

DASRBR010000691.1:1-7214
Verrucomicrobiia bacterium

UDPGP Transp_cyt_pur ParE_toxin ? Amidohydro_1 Aldedh 23S_rRNA_IVP DHO_dh

LCHU01000014.1:6708-15998

Candidatus Giovannonibacteria bacterium GW2011_GWA2_44_13b

DcpS_C tRNA-synt_His tRNA-synt_His HGTP_anticodon Peptidase_S26 Pept_tRNA_hydro RecG_dom3_C Helicase_C DEAD CarD_TRCF_RID UvrB_inter Inositol_P S1 S1 S1 YafQ_toxin RelB ? ? ?

JACCWM010000100.1:1-14782
Lujinxingiaceae bacterium

HATPase_c_3 ? ? Transp_cyt_pur PhdYeFM_antitox ParE_toxin ? Amidohydro_1 Aldedh Aminotran_3 PseudoU_synth_2 S4 AbiEii Rv0078B

JACMLQ010000684.1:580-3661
Anaerolineae bacterium

CopG_antitoxin HTH_3 UPF0150 HicB ParE_toxin PhdYeFM_antitox Mrr_cat Methylase_S

JACMIR010000868.1:1-3776
Fibrella sp.

OEP HlyD_D23 dCMP_cyt_deam_1 ParE_toxin RelB-like RNB S1

JACNKN010000066.1:73214-92259
Candidatus Neomarinimicrobiota bacterium

Bgal_small_N LacZ_4 Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N Aminotran_3 APH Amino_oxidase Amino_oxidase Peptidase_M24 AMP_N ? DUF6290 ParE_toxin PAP2 Orn_Arg_deC_N Glyco_hydro_18 Ig_J_second Por_Secre_tail NamZ_N NamZ_C Peptidase_S13 DUF559 Lactamase_B_2

JACNFU010000339.1:23295-39389
Chloroflexota bacterium

? ELFV_dehydrog_N ELFV_dehydrog CBS CBS ? Melibiase Melibiase_C ? CoA_trans CoA_trans ParE_toxin HTH_3 MazE_antitoxin PIN CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ATP-grasp_5 APH Cass2 HTH_18 Cass2

JACVRQ010000519.1:1-3540
Coleofasciculus sp. Co-bin14

Glycos_transf_1 FTR1 ParE_toxin Uma2

JACVRH010000373.1:2842-5100
Nostoc sp. C3-bin3

? DndB ? ParE_toxin ? ABC_tran_CTD

JACVRH010001430.1:1-1631
Nostoc sp. C3-bin3

Uma2 ParE_toxin ?

JACJQI010000003.1:120470-141334
Aulosira sp. FACHB-113

TPS ? DuOB DUF488 DUF488 PIN bMG10 ParE_toxin ? Big Big bMG3 MG2 bMG5 bMG6 A2M_BRD A2M bMG10 BiPBP_C Transpeptidase Transgly ? Host_attach Receiver Trans_reg_C Hpt Receiver Cupin_2

JACJQI010000052.1:37414-57137
Aulosira sp. FACHB-113

AAA_lid_3 AAA SH3_3 PG_binding_1 PG_binding_1 PG_binding_1 PG_binding_1 PG_binding_1 GAF PAS GGDEF EAL Transglut_core2 TPRs ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N ATP-synt_DE ATP-synt_DE_N ParE_toxin ? ? ? DO-GTPase2 Dynamin_N Band-7

JACSRX010000035.1:70973-82313
Chthoniobacterales bacterium

? ? ? ? ? ? RelB-like ParE_toxin ? ? ? ? ? ? ?

JABTUN010000001.1:16148-31667
Verrucomicrobiaceae bacterium

MFS_1 Glyco_hydro_77 YaeQ CHAD Asp_Glu_race OTCace_N OTCace Aminotran_3 ParE_toxin ? AA_kinase ArgJ Semialdhyde_dh Semialdhyde_dhC_1 ClpB_D2-small AAA_2 ? RecR_C Toprim_4 RecR_HhH

JADEWQ010000078.1:1635-17275
Pleurocapsales cyanobacterium LEGE 06147

? PPDK_N PEP-utilizers PEP-utilizers_C ParA-Soj-NTPase ? ? ? SRAP ParE_toxin RelB-like ? WD40_Prp19 Beta-prop_WDR3_2nd Beta-prop_THOC3 Beta-prop_EML_2 nSTAND1 Caspase DUF899 MFS_1 SnoaL_4 Glyoxalase

JADGCT010000232.1:1-3804
Desulfamplus sp.

? PKD K319L-like_PKD Big RelB-like ParE_toxin Bac_export_2

JADGCQ010000047.1:6428-23412
Magnetococcales bacterium

? ? Hpr_kinase_C Sulfotransfer_3 TPR_CcmH_CycH TPRs NTP_transf_5 Asn_synthase GATase_7 AAA ParE_toxin ? VTT_dom FeoB_associated NusB RsmF-B_ferredox Methyltr_RsmB-F MMR_HSR1 Dus His-Kinase His-Kinase PAS HTH_8 AAA_lid_14 NtrC-AAA Receiver

JACOUS010000031.1:16151-12202
Ignavibacteriales bacterium

Glyco_hydro_92N Glyco_hydro_92 dCMP_cyt_deam_1 ATP-cone PRK PDDEXK_3 RibD_C ParE_toxin PhdYeFM_antitox Pro_CA Peptidase_S15 Glyoxalase_6 His-Kinase His_kinase Por_Secre_tail Glyco_hydro_26 Abhydrolase_2

JACOVD010000200.1:716-6554
Acidobacteria bacterium

Gln-synt_C Gln-synt_N P-II ? DUF5677 ParE_toxin RelB-like GlnD_UR_UTase HD ACT ACT_ACR9_3rd

JACOXK010000068.1:28-10828
Deltaproteobacteria bacterium

GIY-YIG ? Sigma70_r4_2 Sigma70_r2 ACR_tran ACR_tran HlyD_D23 HlyD_D23 HMBD OEP OEP PhdYeFM_antitox ParE_toxin ? Haem_bd Cu-oxidase_2 Cu-oxidase Cu-oxidase_3

JACPAK010000002.1:248459-267181
Deltaproteobacteria bacterium

PhdYeFM_antitox ECH_1 BKACE Carboxyl_trans Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ? ? ParE_toxin RHH_1 ACT_9 ACT_9 AA_kinase LeuA_dimer HCS_D2 HMGL-like OmpA Phenol_MetA_deg RecG_dom3_C Helicase_C DEAD RecG_wedge His-Kinase His-Kinase Cys_Met_Meta_PP

JACPFC010000066.1:16559-37089
Deltaproteobacteria bacterium

DUF2283 AAA_14 DUF4143 ? ? ? ? DUF6290 ParE_toxin DUF4143 AAA_14 NAGPA ATP-synt_ab Rho_RNA_bind Rho_N Lyase_1 ASL_C2 Orn_Arg_deC_N DAP_epimerase DAP_epimerase DHDPS

JACPGD010000365.1:4676-9648
Candidatus Hydrogenedentota bacterium

Cep192_D4 ParE_toxin RelB-like OmpA TPR_NPHP3 TPRs Exonuc_VII_L

JACPPF010000034.1:16369-35069
Planctomycetota bacterium

GFO_IDH_MocA GFO_IDH_MocA_C3 Bact_transglu_N Transglutaminase 3keto-disac_hyd TPRs ? FGGY_N FGGY_C PhdYeFM_antitox ParE_toxin ? DUF4058 Secretin TPRs ? TRAM PIN HD_assoc HD HIT

JACPPF010000114.1:91950-113027
Planctomycetota bacterium

? DUF4175 DUF4175 SQHop_cyclase_N SQHop_cyclase_C Peptidase_S41 PDZ_Tricorn Tricorn_C1 Beta-prop_THOC3 PD40 PD40 ? Uma2 Aldedh ParE_toxin RelB-like DUF4058 Methyltransf_31 AMP-binding DUF6036 ? PQQ_2 PQQ_2

JACPSE010000154.1:210-10653
Deltaproteobacteria bacterium

DUF2442 DUF4160 IspD CarD_TRCF_RID CarD_C Acetylase Fer4 Int_N Phage_integrase ParE_toxin PhdYeFM_antitox HTH_17 ? Zn_ribbon_DnaG HTH_17 ? HTH_17

JACPTQ010000044.1:97430-115332
Deltaproteobacteria bacterium

HTH_3 DUF4011 AAA_11 AAA_11 AAA_12 MTES_1575 REase SWI2_SNF2 T1R_D3-like T1RH-like_C DUF3644 Methylase_S HsdM_N N6_Mtase HTH_31 ParE_toxin PhdYeFM_antitox HTH_11 WYL WCX FHA Prok-RING_1 ? ? ?

JACPVI010000057.1:37478-59540
Myxococcales bacterium

DUF1638 GAF_2 His-Kinase His-Kinase OmpA CarboxypepD_reg CarboxypepD_reg ? Transglutaminase FHA NtrC-AAA AAA_lid_14 HTH_8 DDE_Tnp_1_4 DDE_Tnp_1_4 PhdYeFM_antitox ParE_toxin GxGYxYP_C GxGYxYP_N_3rd GxGYxYP_N_2nd GxGYxYP_N AAA_16 STY-kinase ? OPT STY-kinase DUF3060 DUF3060 ADH_N ADH_zinc_N

JACQSC010000004.1:145129-157334
Candidatus Uhrbacteria bacterium

? Peptidase_S26 ELFV_dehydrog_N ELFV_dehydrog Peptidase_M32 NTP_transferase ? DUF6364 YafQ_toxin ? LAGLIDADG_1 ? LAGLIDADG_3 tRNA_m1G_MT KH_KhpA-B Ribosomal_S16

JACROX010000062.1:209-11997
Candidatus Peregrinibacteria bacterium

? LIM Helicase_C_2 RNase_T AdoHcyase AdoHcyase_NAD S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C AAA_14 DUF4143 HigB-like_toxin RHH_1 dTDP_sugar_isom TauE Epimerase NTP_transferase

JACRFD010000010.1:7316-26620

Ignavibacteriales bacterium

B12-binding Radical_SAM DUF4070 MvaI_BcnI N6_N4_Mtase N6_N4_Mtase MFS_1 MFS_1 CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ATP-grasp_5 Acetylase GatD_N Asparaginase Asparaginase_C GatB_N GatB_Yqey ParE_toxin RelB-like Sortilin-Vps10 Por_Secre_tail Lyase_aromatic YHS HMA Hydrolase E1-E2_ATPase HMA Memo

JACRCX010000047.1:114898-127017
Deltaproteobacteria bacterium

? ? ? MIF Amidohydro_3 Amidohydro_3 ? ? ParE_toxin PhdYeFM_antitox Aminotran_5 ACT PGDH_inter 2-Hacid_dh_C Radical_SAM

JACRCO010000065.1:749-11938
Deltaproteobacteria bacterium

AbiEi_4 TfoX_N AHSA1 Aldo_ket_red Pro_dh PutA_N ORF6N PhdYeFM_antitox ParE_toxin DUF748 DUF748 Vgb_lyase Vgb_lyase DLH ? Glyoxalase_6 adh_short_C2 Isochorismatase

JAENXR010000452.1:422-1880
Deltaproteobacteria bacterium

PhdYeFM_antitox ParE_toxin TfoX_N His_biosynth

JADJOJ010000010.1:653161-667712
Candidatus Leptofilum gracile

COX2 UbiA COX1 Cytochrom_C Cytochrome_B Fer4_7 FlpD NUDIX PhdYeFM_antitox ParE_toxin DUF1638 S-methyl_trans DinB_2 UPF0182 Cys_Met_Meta_PP PNPOx_N MazE_antitoxin

JADJPM010000004.1:3000212-3015960
Verrucomicrobiaceae bacterium

GFO_IDH_MocA GFO_IDH_MocA_C2 LON_substr_bdg AP_endonuc_2 Fer4_13 bPH_2 DUF218 SGL ParE_toxin ? Esterase Methyltr_RsmB-F RsmF-B_ferredox NusB VWA_2 BatA PSD1 PSCyt2 PSCyt1 HXXSHH CMAS

JADJYZ010000006.1:195671-210610
Deltaproteobacteria bacterium

? ? WCX WYL HTH_11 ? ? DUF2237 Alpha-amylase Malt_amylase_C ParE_toxin ? Peptidase_S51 ACP_syn_III ACP_syn_III_C Chromate_transp Chromate_transp Cytochrom_C HYDIN_VesB_CFA65-like_Ig PEGA PEGA

JAEUWE010000360.1:18-1941
Bdellovibrionaceae bacterium

TrbI ParE_toxin RHH_10 ? HTH_17

JAEUVS010000044.1:128-8105
Chitinophagaceae bacterium

? ? ? RelB-like ParE_toxin DUF1508 Aminotran_3 Hexapep Hexapep ? DUF3630 RsgA_GTPase RsgA_N VWA BatA

JAEUUC010000007.1:27511-38650
Cyclobacteriaceae bacterium

? ? ? DUF5684 Lycopene_cyc Lycopene_cyc Cyt_bd_oxida_II Polyketide_cyc2 ParE_toxin RelB-like DNA_photolyase FAD_binding_7 DUF6435 NUDIX SQS_PSY Amino_oxidase Sigma70_r2 Sigma70_r4_2

JAEUQV010000086.1:1-15525
Candidatus Saccharibacteria bacterium

LPG_synthase_TM UbiA COX4_pro COX3 COX1 COX2_TM COX2 COX_ARM ParE_toxin DUF6290 FAD_binding_4 FAD-oxidase_C Pyr_redox_2 PaaX_M PDDEXK_1 UvrD_C UvrD_C UvrD-helicase MIP CDCA

VGFC01000236.1:2769-10590
Armatimonadetes bacterium

Glyco_hydro_42 CBM_4_9 ParE_toxin ? SIS_2 ? ? ? NPCBM_assoc Ig_VWA7 CarboxypepD_reg CarboxypepD_reg ?

JAFGIA010000364.1:20701-29667
Anaerolineae bacterium

Rubrerythrin DUF488 DUF488 DuOB HRDC RQC RecQ_Zn_bind Helicase_C DEAD TPRs ? PhdYeFM_antitox HTH_3 ParE_toxin VWA_2

JAFGTD010000010.1:1-13346
Acidobacteria bacterium

Amino_oxidase DUF169 OPT ParE_toxin DUF6290 Lactamase_B Glyco_hydro_31_2nd Glyco_hydro_31_3rd ? UxaE DctM DctQ

JAFGRW010000100.1:488-11522
Spirochaetes bacterium

? Bax1-I ? RelB-like HTH_19 ParE_toxin RelB-like MCPsignal HAMP sCache_2 ? Big Big MFS_1 Bact_hydrolase AAT TonB_C

JAFGWX010000060.1:10052-17062
Methylothermaceae bacterium

PALP ? DUF1318 TRAM tRNA_U5-meth_tr YkuD UPF0114 RelB-like ParE_toxin GIIM

JAFITR010000011.1:11733-18088

Simkania negevensis

GDH_HM3 Bac_GDH_CD Bac_GDH_CD ELFV_dehydrog Hydrolase E1-E2_ATPase VIT1 ? ParE_toxin GDC-P GDC-P GcvP2_C Aconitase Aconitase_C LicD ? Glyco_transf_9 TLC

JAAVVT010000003.1:2521597-2533832
Dolichospermum sp. DEX182a

PDDEXK_3 GIY-YIG ? PemK_toxin HicB_lk_antitox HicA_toxin DUF4276 ParE_toxin ? NAD_binding_2 6PGD Collagen Ycf66_N AAA_26 DRTGG LAM_C Radical_SAM Dala_Dala_lig_C

RFFC02000037.1:47185-64876
Hormoscilla sp. GUM202

? Beta-Roll Beta-Roll LRR_14 LRRs He_PIG LRR_14 LRR_14 LRR_14 LRRs LRRs LRR_14 Peptidase_M10_C Beta-Roll LRR_14 LRRs LRR_14 LRR_14 He_PIG LRR_14 LRRs LRR_14 LRR_14 TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C Zn_Ribbon_1 Uma2 ? PhdYeFM_antitox ParE_toxin PhdYeFM_antitox PhdYeFM_antitox Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans DUF2281 PemK_toxin ? ?

JAFNIZ010000035.1:3351-8383
Candidatus Brockarchaeota archaeon

HicB_lk_antitox ? ? ? EVE ? ? ParE_toxin UPF0175 PIN DUF433 ?

JAFOEX010000001.1:260-3676
Spirochaetaceae bacterium

DUF6290 ParE_toxin Replicase PriCT_1 HTH_23 Mob_Pre DUF6290 ParE_toxin

JAANWY010000387.1:710-7069
Anaerolineales bacterium

APH ParE_toxin PhdYeFM_antitox UPF0029 DUF1949 TPP_enzyme_C Fer4_7 EKR POR PFOR_II POR_N

JAFDXN010000012.1:135234-143161
Bacteroidetes bacterium

? Por_Secre_tail DUF6089 NAD_kinase NAD_kinase_C Beta-prop_EML_2 Cytochrom_D1 PSCyt1 FolB FGAR-AT_linker AIRS AIRS_C AIRS AIRS_C ParE_toxin RelB-like ?

JAGVYZ010000020.1:11020-22739
Candidatus Woesearchaeota archaeon

Fic KilA-N ? ? Yip1 ? DUF7557 ParE_toxin tRNA_SAD tRNA-synt_2b HGTP_anticodon ? ? FmrO ? Flavin_Reduct S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C

JAGXRH010000024.1:28911-45274
Ignavibacteriales bacterium

GIDA GIDA_C_1st SAM_GIDA_C TrmE_N MnmE_helical Lip_A_acyltrans Glyco_transf_9 ABC_membrane ABC_tran RtcB TMEM175 ParE_toxin RelB-like ? ? RE_TdeIII N6_N4_Mtase Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 Fic

JABJOR010000579.1:1-3919
Rhodospirillales bacterium

MTTB MTTB HEPN_AbiU2 PhdYeFM_antitox ParE_toxin

JAHHMP010000021.1:7770-22027
Deltaproteobacteria bacterium

DUF1330 Bac_luciferase adh_short ADH_N ADH_zinc_N_2 DUF4437 ? DUF6290 ParE_toxin AA_permease_2 TrkA_N TrkA_C p450 TetR_N Rieske KshA_C MGH1-like_GH MGH1-like_GH

JAHJDR010000326.1:1736-3509
bacterium

? AbiEi_2 ParE_toxin DUF6290 Pribosyltran Peptidase_M23

JAHJOV010000038.1:120-4916
Patescibacteria group bacterium

RHH_1 YafQ_toxin RuvB_C AAA_lid_4 RuvB_N T2SSE MshEN Peptidase_S24

JAHJOU010000072.1:3097-7481
Bacteroidetes bacterium

CoA_transf_3 ParE_toxin RelB-like HMGL-like Iso_dh POR TPP_enzyme_C POR_N PFOR_II Fer4_7 Fer4_16

JAHJTI010000012.1:1-11129
Proteobacteria bacterium

? TPRs ParE_toxin PhdYeFM_antitox DUF6172 ParE_toxin Antitox_RHH DUF4268 REase SWI2_SNF2 T1R_D3-like T1RH-like_C Methylase_S Methylase_S HsdM_N N6_Mtase Phage_CI_repr

JAHIFQ010000002.1:2736-15218
Patescibacteria group bacterium

Peptidase_C1 Peptidase_C1 T4SS_pilin PolyA_pol PolyA_pol_RNAbd HD NMD3 Spermine_synth DUF6364 YafQ_toxin DUF192 UPF0102 NusA_N S1 KH_5 KH_NusA_C H_PPase Trigger_N Trigger_C CLP_protease RNase_Y_N KH_1 HD

JAFCZJ010000240.1:1-5849
Deltaproteobacteria bacterium

OmpA ? ParE_toxin Calcineurin DUF2279 TetR_N Peptidase_M15 TPRs TPRs Receiver MshEN

JAHYJK010000008.1:35254-50603
Anaerolineaceae bacterium

MFS_2 Radical_SAM AXE1 N_BRCA1_IG ? Peptidase_M24 HTH_3 ParE_toxin ABC_tran ABC_tran BCA_ABC_TP_C BPD_transp_2 BPD_transp_2 Peripla_BP_6 Peptidase_S8 Inhibitor_I9

JAHBUR010000012.1:73309-89451
Bacteroidetes bacterium

Abhydrolase_1 Calcineurin Na_Pi_cotrans Na_Pi_cotrans MS_channel_1st MS_channel_2nd MS_channel_3rd CitMHS CitMHS LapA_dom Radical_SAM CofH_C ParE_toxin PhdYeFM_antitox Radical_SAM CofH_C VitK2_biosynth ? ? Pan3_PK AAA_23 ORF6N

JAHBTI010000010.1:54923-69893
Bacteroidetes bacterium

VitK2_biosynth PNPase DUF1015 2-Hacid_dh_C GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C HTH_26 HKD ParE_toxin ? ? QueG_DUF1730 Fer4_16 PHA_synth_III_E Abhydrolase_1 POTRA POTRA ShlB Por_Secre_tail

JAHBWO010000055.1:6770-19601
Nitrospira sp.

Sua5_C Sua5_yciO_yrdC Transposase_20 DEDD_Tnp_IS110 DUF6982 ODP Bestrophin-like TPRs TPRs ScsC_N DSBA ParE_toxin PhdYeFM_antitox TPM_phosphatase ? FAD_binding_3 HKD ? ThiS

JAIERW010000021.1:2460-12244
Candidatus Babeliales bacterium

GIY-YIG RlaP ? Acetylase TfoX_N AhpC-TSA PhdYeFM_antitox HTH_3 ParE_toxin APH Peptidase_M48 Abhydrolase_3 DUF2208 ? Peptidase_S26 GIY-YIG

JAIEUA010000010.1:422111-436979
Phycisphaerales bacterium

? ? T2SSG N_methyl T2SSG T2SSG T2SSF T2SSF PhdYeFM_antitox ParE_toxin ADH_N ADH_zinc_N ? ? ? TPRs STY-kinase TetR_N CecR_C MacB_PCD FtsX

JAIOHW010000998.1:1-2128
bacterium

ParE_toxin PhdYeFM_antitox MIP Apiosidase Collagen_bind_2

JAIQGH010000017.1:4252-18279
Terriglobia bacterium

UnbV_ASPIC FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 HZS_alpha HTH_Crp_2 cNMPBD Citrate_synt tRNA_edit DUF2461 RelB-like ParE_toxin Sigma70_r2 Sigma70_r4_2 ? DUF3106 Methyltransf_25 Trypsin PDZ_2 ? DUF3303

JAIQGH010000149.1:5528-7893
Terriglobia bacterium

? ? UbiD_N UbiD UbiD_C PhdYeFM_antitox ParE_toxin ? DDE_Tnp_IS66_C ?

JAIQGA010000082.1:1-11849
Terriglobia bacterium

DDE_Tnp_1 ? ParE_toxin PhdYeFM_antitox Rhodanese Rhodanese TPRs TPRs TPRs TPRs ? TMP-TENI Lyase_1 ADSL_C Lactamase_B_2

JAIQFW010000066.1:16406-22264
Terriglobia bacterium

? ? ? 2-oxoacid_dh E3_binding Biotin_lipoyl Cytochrome_CBB3 Transketolase_C Transket_pyr PhdYeFM_antitox ParE_toxin E1_dh

JAGQGS010000001.1:11426-18161
Thermomicrobiales bacterium

adh_short_C2 ? Uma2 Rieske Amidohydro_2 YjgF_endoribonc Aminotran_5 ParE_toxin PhdYeFM_antitox PhdYeFM_antitox

JAGQXP010000278.1:1425-13361
Phaeodactylibacter sp.

? ? YHS CarboxypepD_reg Plug TonB_dep_Rec_b-barrel ? NTPase_1 ? ParE_toxin ? Trans_reg_C Receiver His-Kinase

JACKEV010000008.1:705356-720207
Alphaproteobacteria bacterium

Peptidase_S9 Peptidase_S9_N Zn_ribbon_TFIIB Zn_ribbon_TFIIB Usp Creatininase Usp ? Cu-binding_MopE Epimerase Cupin_2 ? ParE_toxin ? Lambda_Bor FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP_3 FG-GAP_3 MASE1 His-Kinase His-Kinase Acetylase ? ?

JAJHQX010000020.1:21514-28610
Armatimonadota bacterium

Methyltransf_4 Uma2 Amidase TPK_catalytic SteA-like_C MctB AAA_3 AAA_lid_2 ? ParE_toxin

JADLCK010000068.1:57773-8141
Anaerolineales bacterium

? Methyltransf_31 HepT-like Polbeta ParE_toxin PhdYeFM_antitox PIN PhdYeFM_antitox ? DUF3368 UPF0175 Mur_ligase_C Mur_ligase_M

JADLAP010000119.1:6760-14132
Phycisphaeraceae bacterium

? PhdYeFM_antitox ParE_toxin LacI Peripla_BP_3 N_methyl ? DUF4838 CBM9_1 ? GH123_N GH123_N LamininG LamininG

JAGHCU010000094.1:3195-4914
Candidatus Latescibacterota bacterium

FlgD_ig Glyco_transf_9 ? ParE_toxin NTP_transf_2 ? ?

QWGW01000019.1:23410-39738
Haliscomenobacteraceae bacterium CHB4

? TPRs Sigma70_r4 Sigma70_r2 OmpA DUF937 ? MRM3-like_sub_bind SpoU_methylase MazG MazG ParE_toxin Ribosomal_S15 ? RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 PKD_4 Por_Secre_tail ? RoxA-like_Cyt-c CCP_MauG SnoaL

JAKSAX010000221.1:208-685
Desulfobacterales bacterium

ParE_toxin RH3_dom

JAKRSB010000040.1:16047-15963
Thermosynechococcaceae cyanobacterium MS004

HTH_24 SufBD_N SUFBD_core Uma2 Transglutaminase Ig-like_KY ABC_tran SufBD_N SUFBD_core PhdYeFM_antitox ParE_toxin Aminotran_5 PAD_porph CN_hydrolase LPG_synthase_TM Pps Pps SWIM TP_methylase CbiG_C CbiG_N

JAFLSQ010000019.1:1249-8449
Treponema sp.

? ParE_toxin DUF6290 Ribosomal_S21 GntR FCD Cupin_2 HTH_18 ABC_tran ABC_membrane ABC_tran ABC_membrane GH43_C2 Glyco_hydro_43

JADFQN010000138.1:1-8323
Bacteroidetes bacterium

ShlB Lactamase_B_2 Methyltransf_25 TauE OsmC Peptidase_M20 MTHFR ParE_toxin RelB-like MTHFR ? COX4_pro

JACZRV010000007.1:17317-31635
SAR324 cluster bacterium

? FCSD_central Pyr_redox_2 ? HicB_lk_antitox B12-binding MeaB MM_CoA_mutase MM_CoA_mutase DJ-1_PfpI MerR_1 MerR-DNA-bind ParE_toxin PhdYeFM_antitox ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C OSCP ATP-synt_B ATP-synt_B

JAKEFM010000023.1:1-11067
Planctomycetota bacterium

Fic PhdYeFM_antitox PIN Peptidase_M64 M64_N ParE_toxin PhdYeFM_antitox SLBB_2 SLBB Poly_export Glycos_transf_2 Glycos_transf_2 Glycos_transf_1 Glyco_transf_4 ? PMT_2

JAKEFD010000147.1:1-10457
Gemmataceae bacterium

STY-kinase HD_5 GAF_2 FHA DUF4139 ParE_toxin ? AP_endonuc_2 ? ? Peptidase_M48 Peptidase_M48_N Polysacc_deac_1

JALHTC010000249.1:1-1954
Anaerolineales bacterium

NNH1 ParE_toxin HTH_3 DUF559

JALHUF010000027.1:1-4525

Terriglobia bacterium

BipA_C EFG_C EFG_III GTP_EFTU_D2 ? HTH_3 HigB-like_toxin ParE_toxin ? ?

JALHWK010000648.1:1-9110
Sedimentisphaerales bacterium

ParB_N HTH_Tnp_1_2 LacAB_rpiB PhdYeFM_antitox ParE_toxin Terminase_6C HTH_3 Peptidase_M78 ? Hydrolase_4 GCE_fung ?

JALHYD010000467.1:742-4291
bacterium

HicA_toxin HicB_lk_antitox ? DUF7387 HicA_toxin ? RelB-like ParE_toxin Caspase

JAGLZM010000205.1:355-6397
Candidatus Aegiribacteria sp.

Yip1 ? MgtE_N CBS CBS MgtE GlgB_N CBM_48 Alpha-amylase Alpha-amylase Alpha-amylase_C ? RelB-like ParE_toxin

JAGLZA010000040.1:958-4794
candidate division WOR-3 bacterium

His_Phos_1 BPL_C BPL_LplA_LipB Peptidase_M48 Peptidase_M48_N RelB-like ParE_toxin

JAGLZA010000201.1:4329-17517
candidate division WOR-3 bacterium

TEN_YD-shell ? ? F-box PIN ? RelB-like ParE_toxin RHS DUF192 DeoC SMC_N CHASE2 Guanylate_cyc B12-binding Radical_SAM NurA

JAGLTD010000388.1:1-4874
Spirochaetes bacterium

Cas_Cas7 Cas_TM1802 Polbeta ParE_toxin RelB-like Cas6b_N Cas6b_C CBM_6

JAIQZM010000030.1:30945-45699
Argonema galeatum A003/A1

Peptidase_M23 Fer4_5 Fer4_5 AAA_lid_14 NtrC-AAA cNMPBD DUF488 DUF488 DuOB ? ? ParE_toxin PhdYeFM_antitox N6_N4_Mtase ? ? Uma2 PhdYeFM_antitox Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans

JAMCTI010000016.1:23319-31779
Patescibacteria group bacterium

THF_DHG_CYH_C THF_DHG_CYH PhdYeFM_antitox DUF2281 PIN PIN DUF6364 AAA_33 RelB YafQ_toxin LMWPc ? PEMT YgaP-like_TM Glutaredoxin FUR ?

JAMCWE010000265.1:13123-23748
Bacteroidetes bacterium

GWxTD_dom 2-Hacid_dh_C PGDH_inter ACT Thioredoxin adh_short_C2 DUF6883 DUF4926 PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB ParE_toxin RelB-like Polbeta ?

JAMDDR010000562.1:1-6085
Chloroflexota bacterium

? ? ParE_toxin ? HIRAN TM2 Phage_integrase ? Glycos_transf_2 DUF5826 Pribosyltran DZR_2

JAGTJV010000004.1:21802-32677
Gloeotrichia echinulata DEX184

DUF2281 PIN Uma2 CPSF100_C Uma2 DUF29 ? DUF2281 ParE_toxin DUF2281 PIN HTH_Tnp_4 DDE_Tnp_4 Glycos_transf_2 Acyl_transf_3 ? ?

JAGTJX010000002.1:425460-437338
Lysobacteraceae bacterium

LptE DciA ? ABC_tran Ala_racemase_C Ala_racemase_N ? DUF6290 ParE_toxin FAD_binding_2 Succ_DH_flav_C Trans_reg_C Receiver ? ? HD_5 ? SpoIIE

JAAEQP010000826.1:3343-16903
bacterium

GFO_IDH_MocA DUF362 Methyltransf_25 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PQQ_2 PhoD_N DUF2341 LamininG Hydrolase_6 Hydrolase_like HAD_2 PhdYeFM_antitox ParE_toxin UvrD_C UvrD_C UvrD-helicase BD-FAE

JALOOZ010000146.1:1-11550
Thermodesulfobacteriota bacterium

? HlyD_D23 ABC_tran FtsX FtsX MacB_PCD DUF169 ParE_toxin PhdYeFM_antitox MqsA_antitoxin MqsR_toxin CoA_binding_2 Succ_CoA_lig Ligase_CoA_2 ETF ETF_alpha

JAPHTV010000196.1:1-3182
Candidatus Pacearchaeota archaeon

Helicase_C UvrB_inter ResIII DUF7557 ParE_toxin TerC ?

JAPLOD010000808.1:1-7542
Planctomycetota bacterium

Beta-prop_ACSF4 PQQ_2 PQQ_2 TPRs zf-HC2 Sigma70_r2 Sigma70_r4_2 ? ParE_toxin PhdYeFM_antitox SRAP

JAQCNZ010000020.1:151015-159956
Chloroflexota bacterium

ABC_tran TOBE_2 BPD_transp_1 BPD_transp_1 SBP_bac_8 23S_rRNA_IVP Aminotran_3 ? ParE_toxin ? Lysyl_oxidase

JALLOE010000029.1:3011-11710

Nitrospiraceae bacterium AH_259_D15_M11_P09

GerE Receiver GerE PAS PemK_toxin RHH_1 PIN ? BrnT_toxin ParE_toxin MazE_antitoxin ? YqeY ? ? ? ?

JAQUDK010000001.1:15660-25400
Candidatus Gracilibacteria bacterium

Herpes_Helicase SH3_13 PIF1 Lactamase_B Thymidylate_kin Ribosomal_L31 HicA_toxin HicB_lk_antitox tRNA-synt_2 N-OB ParE_toxin ? N_methyl Peptidase_M23

JAQTNW010000051.1:1-8188
Candidatus Methylacidiphilales bacterium

GTP_EFTU_D2 GTP_EFTU Ribosomal_S7 Ribosom_S12_S23 RelB-like ParE_toxin Cupin_2 FumaraseC_C Lyase_1 YceI MotA_ExbB ExbD Pep_deformylase PDDEXK_3

JAQUQV010000013.1:19571-38944
Kiritimatiellae bacterium

tRNA-synt_1b CBM_48 Alpha-amylase Alpha-amylase_C AMPK1_CBM Amidohydro_1 ? NAD_binding_2 6PGD DUF4838 CBM_4_9 ParE_toxin PhdYeFM_antitox TPRs POR TPP_enzyme_C POR_N PFOR_II Fer4_7 ADH_N ADH_zinc_N

JAPWKE010000003.1:2905946-2923168
Acidobacteriaceae bacterium

Aminotran_3 AIRS_C AIRS Formyl_trans_N Peptidase_M50 Peptidase_M50 MerR_1 Vgb_lyase RtcB ParE_toxin ? EF_TS Ribosomal_S2 Ribosomal_S9 Ribosomal_L13 CarboxypepD_reg OMP_b-brl_4 OMP_b-brl_4 TonB_C

JAQSWM010000005.1:3113-21508
Sphingobacteriaceae bacterium

? Peptidase_M19 RibD_C ADH_N ADH_zinc_N ? ? RelB-like ParE_toxin Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind UPF0102 Big Glu_cyclase_2 Bro-N PDDEXK_1 DUF5009 Peptidase_M16 Peptidase_M16_C Peptidase_M16_C

JAOTYS010000732.1:87-2064
Gammaproteobacteria bacterium

RHH_1 DUF6290 ParE_toxin Ysc84 ?

JASJDW010000001.1:17146-31227
Crocosphaera sp.

GST_C_2 GST_N_2 DUF6887 DUF6888 FitA-like_RHH ? PIN HicB_lk_antitox ParE_toxin PhdYeFM_antitox GDE_N GDE_C Peptidase_M16_C Peptidase_M16 Peptidase_M16 Receiver His-Kinase His-Kinase GAF PAS PAS GAF CBS CBS CBS CBS Coprogen_oxidas ?

JASJDN010000035.1:23210-32756
Crocosphaera sp.

Y1_Tnp HAD_2 QueG_DUF1730 Fer4_16 HepT-like NTP_transf_2 Glyco_transf_21 ? ParE_toxin RelB-like CopG_antitoxin Uma2 DUF3226 AAA_15 PIN DUF7082

JASJDN010000044.1:4317-15316
Crocosphaera sp.

SocA_Panacea STY-kinase HEATs HEATs HEATs HEATs HEATs HTH_3 ParE_toxin CTP_synth_N GATase PDZ_2 Trypsin Peptidase_M15 ? Peptidase_M20 DUF2267 SQHop_cyclase_N SQHop_cyclase_C

JASJCT010000022.1:172960-192690
Myxococcota bacterium

Patatin PC4 AIRS_C AIRS AIRS_C AIRS FGAR-AT_linker GATase_5 HTH_17 ABC-ATPase CCDC113_CCDC96_CC AAA_15 HATPase_c_4 SLFN_AlbA_2 ParE_toxin PhdYeFM_antitox ? Ribonucleas_3_3 dsrm Thioredoxin_4 VKOR GreA_GreB GreA_GreB_N RecG_dom3_C Helicase_C DEAD RecG_wedge HEATs SGSH_C Sulfatase

JASJBX010000119.1:12882-5501
Crocosphaera sp.

HTH_37 HicB_lk_antitox ? ? ? ParE_toxin ? ? PaaA_PaaC Uma2 Host_attach Cu-oxidase_4 DUF6887

JAUSEU010000418.1:1231-7952
Dehalococcoidia bacterium

AdoHcyase Helicase_C ? BrxC_BrxD BrxC_BrxD BrxC_BrxD ? ParE_toxin ? SBP_bac_6

JAUXTT010000042.1:172803-185686
Bacteroidota bacterium

PhoH SAM_HAT_N SAM_HAT_C ABM ? HigB_toxin HTH_3 DUF3052 ParE_toxin ? fn3 Por_Secre_tail Glyco_hydro_130 ? ZapA RNase_Y_N KH_1 HD Por_Secre_tail

JAUYMY010000060.1:63010-76093
Methylobacter sp.

DCTN5 cNMPBD HDOD LptF_LptG LptF_LptG Peptidase_M17_N Peptidase_M17 DNA_pol3_chi DUF6290 ParE_toxin GmrSD_N HEPN_RiboL-PSP ? ? ? PIN tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C

JALZKH010000053.1:240-8016
Acidobacteriota bacterium

RelB-like ParE_toxin DUF4365 DUF5678 ? FMO-like NAD_binding_8 E1_dh Na_H_antiport_1 ?

JALZIH010000398.1:1-4809
Acidobacteriota bacterium

? ? ? ParE_toxin ? Peptidase_M15 EPSP_synthase SKI

JALZIH010000441.1:1-3749
Acidobacteriota bacterium

YidC_periplas YidD Ribonuclease_P Ribosomal_L34 ParE_toxin ? Formyl_trans_N CarboxypepD_reg ?

JAIROK010000015.1:1-6109
Campylobacteraceae bacterium

RHH_1 ParE_toxin ? ? YafQ_toxin Relaxase TrbC VirB3 CagE_TrbE_VirB AAA_10

JAISJO010000023.1:1-8626
Propionibacteriaceae bacterium

BRCT ParE_toxin PhdYeFM_antitox Phage_integrase SBP_bac_8 Alpha-amylase ABC_tran DrrA1-3_C ? Zn_ribbon_9 CT398_CC

JAISAC010000193.1:2128-8326
Prevotellaceae bacterium

OEP OEP DUF4861 GATase_2 GATase_6 DUF4160 DUF2442 DUF3802 ? ParE_toxin

JAITCS010000155.1:1135-6045
Cellulomonadaceae bacterium

FAD_syn ? ParE_toxin PhdYeFM_antitox GerE ? ?

JAITOZ010000260.1:1-7922
Planctomycetaceae bacterium

T2SSE FMN_red Beta-prop_THOC3 SHD1 DUF7382 ? ParE_toxin ? DUF4139

JAVVGO010000158.1:71-5089
Armatimonadota bacterium

TatA_B_E Shikimate_DH Shikimate_dh_N PCRF RF-1 2Fe-2S_thioredx ParE_toxin ? Complex1_49kDa Complex1_30kDa

JAVVHJ010000032.1:408-7324
Armatimonadota bacterium

? PBP 2Fe-2S_thioredx ParE_toxin DUF7342 ParE_toxin ? Complex1_49kDa Complex1_30kDa ? HEPN

JANRLM010000031.1:47614-53933
Rickettsiales bacterium

ATP-synt_B ATP-synt_C ATP-synt_A ATPase_gene1 Thioredoxin_4 DciA tRNA-synt_1c Anticodon_2 ParE_toxin HTH_3 tRNA-synt_1e

JAWXCB010000008.1:92126-104462
Chitinophagales bacterium

TetR_C_24 TetR_N PaaA_PaaC PaaB PaaA_PaaC Zn_ribbon_PaaD FeS_assembly_P ? DUF374 ParE_toxin RelB-like DAO DAO_C GIY-YIG ADH_N ADH_zinc_N ADI DDAH_eukar Synaptojanin

JAVRCF010000002.1:170289-185172
Aulosira sp. DedQUE10

Acetyltransf_7 GST_C GST_N_3 AhpC-TSA 1-cysPrx_C AhpC-TSA GDE_N GDE_C Peptidase_M24 ? ParE_toxin Cupredoxin_1 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Peptidase_M16 Glyoxalase DUF2396 Hexapep Hexapep PsbY Mur_ligase_M Mur_ligase_C

JAYEKO010000020.1:11097-20585
Campylobacterota bacterium

Ribosomal_S30AE UPF0093 KH_dom-like MMR_HSR1 MMR_HSR1 LPP20 ? LPP20 Pribosyltran ParE_toxin ? N_methyl N_methyl DUF4258 DUF2283 Fic ?

JAYKJI010000224.1:725-7592
Snowella sp.

WYL WCX Cas6 Cas3_HD DEAD Cas3-like_C_2 ? DevR RHH_4 ParE_toxin Cas_Cas5d

JAYKJI010000431.1:5001-15257
Snowella sp.

STY-kinase FGS 23S_rRNA_IVP ? ? BrnT_toxin BrnA_antitoxin AIPR ParE_toxin DUF2281 XisI STY-kinase FGS Fer4_12 ?

JAYKHF010000030.1:3000-12623
Leptolyngbya sp.

GatC DUF7620 Uma2 Glycos_trans_3N Glycos_transf_3 HTH_1 LysR_substrate Phosphorylase ADIP DDE_Tnp_ISAZ013 ParE_toxin

JAYKGY010000050.1:932-7709
Snowella sp.

Cas3_HD DEAD Cas3-like_C_2 ? BrnT_toxin Cas8b_N Cas8b_C DevR Cas_Cas5d RHH_1 ParE_toxin

JAWQFP010000179.1:4547-4383
Scytonema sp. PMC 1069.18

PIN Uma2 ? ParE_toxin ? Calcineurin ? ?

QGUM02000135.1:1-4491
Actinomycetes bacterium

Sod_Fe_C Sod_Fe_N ParE_toxin PhdYeFM_antitox adh_short HSDR_N_2 Cyt_c_ox_IV COX1

WJFC01000001.1:257368-273595
Scytonema sp. UIC 10036

? TPS PIN ? ? TPRs TPRs TPRs RNaseH-CATRA+Caspase ? ParE_toxin DUF928 XisI Antibiotic_NAT Fer2 Fer2_2 FAD_binding_5 CO_deh_flav_C Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 Coq4

WJFC01000015.1:85327-97219
Scytonema sp. UIC 10036

MCPsignal HAMP TPR_CcmH_CycH CheR CheR_N CheW ? ? ? ? ParE_toxin ? DUF6972 DUF1016_N DUF1016_N YhcG_C Gp49 ? His-Kinase Uma2

WJFC01000025.1:32204-48682
Scytonema sp. UIC 10036

DUF1003 Rrf2 TauD Acyl-CoA_dh_2 Acyl-CoA_dh_M Acyl-CoA_dh_N Bac_luciferase Acyl-CoA_dh_2 Acyl-CoA_dh_M Acyl-CoA_dh_N UPF0181 ParE_toxin ? Bac_luciferase ABC_membrane ABC_tran Aldo_ket_red Methyltransf_25 Abhydrolase_2 Pyr_redox_3

VYEN01000136.1:6447-6369
Gemmatimonadetes bacterium

DUF5128 DinB_2 DUF4983 Phosphodiest ? ? RHH_1 ParE_toxin

JAAAOF010000278.1:48-7279
Bacteroidetes bacterium

Receiver HTH_18 Receiver His-Kinase His-Kinase Y_Y_Y Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop Reg_prop ParE_toxin PhdYeFM_antitox Phage_integrase

JAACPR010000006.1:21910-39067
Candidatus Parcubacteria bacterium

Sulphotransf Sulphotransf ? Glycos_transf_2 TatD_DNase tRNA-synt_1g tRNA-synt_1g Anticodon_3 AI-2E_transport Peptidase_M16 Peptidase_M16_C ParE-like_toxin RHH_1 tRNA-synt_2b HGTP_anticodon DUF2207_C DUF11 Adap_comp_sub RecO_N tRNA-synt_1 His_Phos_1 tRNA-synt_1 Anticodon_1 DUF5915 ? ?

JAAHGY010000434.1:42-5924
Cyanothece sp. SIO2G6

? PhdYeFM_antitox ParE_toxin PPC TPRs TPR_NPHP3 TPR_MalT TPRs TPRs TPRs TPRs TPRs RNaseH-CATRA+Caspase

JAAKFX010000055.1:1-10326
Candidatus Anoxychlamydiales bacterium

HTH_16 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7 RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 MFS_1 ? TAL_FSA PhdYeFM_antitox ParE_toxin ?

JAACEL010000445.1:197-3689
Nitrospiraceae bacterium

peroxidase peroxidase ParE_toxin RelB-like

JAAUTN010000243.1:1-5656

Leptolyngbyaceae cyanobacterium SM1_4_3

TPRs TPRs ? ParE_toxin UPF0150 DUF2283 DAO DAO FA_desaturase ? EAL GGDEF

JAAUUQ010000134.1:5058-6680
bacterium

NACHT HEATs RelB_like_antitoxin ? ? ParE-like_toxin DUF2281 ParE_toxin CTP_transf_1 RRF DDE_Tnp_IS1

JAAUPN010000074.1:5944-17165

Coleofasciculaceae cyanobacterium RL_1_1

? SpoIIE Arginase MethyltransfD12 MethyltransfD12 DUF3747 Proton_antipo_N Proton_antipo_M Proton_antipo_C ParE_toxin DUF2281 Hexapep

JAAUPR010000046.1:17041-28729

Limnothrix sp. RL_2_0

? Nucleos_tra2_N Gate Nucleos_tra2_C MNS-STAND Beta-prop_THOC3 Beta-prop_THOC3 Beta-prop_WDR3_1st MNS-STAND GGDEF MNS-STAND DUF4335 DUF3379 DUF3038 ParE_toxin DUF2281 GIY-YIG ?

JABFSO010000050.1:1-7949
Cyclobacteriaceae bacterium

Peptidase_M3 Peptidase_M3 ParE_toxin RelB-like MarR YceI CBM9_1 DUF5916 MTS Fe-ADH ADH_Fe_C AbiEii AbiEii

JAADGV010000099.1:12947-18606
Planctomycetota bacterium

HTH_17 ? DUF669 DnaB_C HTH_24 ? PhdYeFM_antitox ParE_toxin Phage_int_SAM_5 Phage_integrase

JABMMJ010000277.1:1-2860
bacterium

Ribosomal_L27 OpcA_G6PD_assem OpcA_G6PD_C ParE_toxin RelB-like G6PD_C

JABUFB010000009.1:223317-243948
Armatimonadetes bacterium

Put_Phosphatase PMT Hepar_II_III_C Hepar_II_III_N CBM_11 GFO_IDH_MocA_C3 GFO_IDH_MocA ? Acetylase Rab11-FIP3 Zn_Ribbon_1 ParE_toxin DUF5696 Amidohydro_2 3keto-disac_hyd BNR_2 Amidohydro_2 PAP2 CBM9_1 Glyco_hydro_20b Glyco_hydro_20 DUF3429

JAATWA010000001.1:97086-109656
Pseudanabaena biceps

AAA_23 Rad50_zn_hook Tup_N SbcC_Walker_B PemK_toxin ? TPRs TPRs TPRs STY-kinase ? ? ParE_toxin Radical_SAM LIAS_N ? DUF3318 TPRs TPRs TPRs TPRs Zn_ribbon_LapB ATPase ChlI ? NinF PALP

MEKY01000062.1:15241-12654

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_59_11

? ? 5_3_exonuc_N 5_3_exonuc DNA_polI_exo1 DNA_pol_A PIN PhdYeFM_antitox ParE_toxin LolA_like ? ABC_tran FtsX MacB_PCD LolA_like ? CarboxypepD_reg Plug OMP_b-brl_4

MGBU01000112.1:9275-9273

Candidatus Rokubacteria bacterium GWA2_73_35

ParE_toxin ? DUF6941 EamA EamA AFOR_N AFOR_C TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs Hydrolase

MGRG01000008.1:186274-200671

Deltaproteobacteria bacterium RIFCSPHIGHO2_01_FULL_43_49

T2SSE MshEN POR POR_N TPP_enzyme_C PFO_beta_C CSD Omp85 CN_hydrolase DUF6290 ParE_toxin ? Hydrolase_4 DUF433 DUF5615 KARI_C KARI_N ? Anticodon_2 tRNA-synt_1c

MGRR01000041.1:8200-7364

Deltaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_43_9

? PhdYeFM_antitox ParE_toxin Transketolase_N Transket_pyr Transketolase_C dnstrm_HI1420 Gp49 Lipase_GDSL ThiI THUMP ThiI_fer Aminotran_5

MGST01000313.1:16361-28618

Deltaproteobacteria bacterium RIFOXYA12_FULL_58_15

AbiEii SNase LTD SNase Fer4_20 Pyr_redox_2 DHO_dh Fer4_21 His-Kinase PilZ ? ParE_toxin XPB_DRD ResIII Helicase_C DUF790 PPDK_N PhdYeFM_antitox

MHXR01000049.1:53153-67983

Planctomycetes bacterium GWA2_50_13

CN_hydrolase NAD_synthase ? HD ? Cas_Cas1 CRISPR_Cas2 ? ParE_toxin Acyltransferase S1 S1 S1 S1 S1 S1 Sigma70_r4 Sigma70_r2 Sigma70_r1_2 Pribosyltran Peptidase_S24 HTH_3 Amidohydro_3 Cytidylate_kin

LWDK01000048.1:38915-51134

Proteobacteria bacterium SG_bin7

ICMT ? ? ? ? RWP-RK Recombinase PAD ParE_toxin RH3_dom PSCyt1 YceI ? ? ? CRCB

NBMI01000049.1:27722-44302
Candidatus Parabeggiatoa sp. nov. 2

RuBisCO_small CbiA LysM SH3_3 ? ? ? ParE_toxin ? ? AAA_15 DUF4435 HSP70 Nuf2_DHR10-like Fer4_12 FGS

PFPS01000054.1:1-2741

bacterium (Candidatus Gribaldobacteria) CG_4_10_14_0_2_um_filter_36_18

? YafQ_toxin Cyclophil_like Inositol_P ? ?

PVWL01000028.1:45416-55668
Cyanosarcina cf. burmensis CCALA 770

Xre_MbcA_ParS_C Xre-like-HTH RES ? HTH_23 ? ? RHH_1 ParE_toxin PhdYeFM_antitox HTH_26 ? Polbeta HEPN ? ?

PXSZ01000044.1:73375-82073

Bacteroidetes bacterium QH_10_64_37

Proton_antipo_M Oxidored_q5_N Pterin_4a Proton_antipo_M ? Aldedh Isochorismatase HicB ParE_toxin RelB-like DUF5615 DUF433 DUF433

PXTJ01000034.1:11230-8822

Bacteroidetes bacterium QH_9_67_14

Peptidase_M48 PhdYeFM_antitox ParE_toxin Amidohydro_1 Metallophos_2 Adenylsucc_synt STAS SecD_SecF_C Sec_GG SecD_SecF_C SecDF_P1_head ?

QNEL01000080.1:216-20854
Candidatus Parabeggiatoa sp. nov. 3

HicB_lk_antitox Glycos_transf_2 Glycos_transf_2 Sulfotransfer_3 ? ? ParE_toxin Calx-beta PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 SPASM Radical_SAM Sulfotransfer_3 Sulfotransfer_1 Sulfotransfer_3 Glycos_transf_2 ? Sulfotransfer_3 AAA_23 ABC-ATPase HicA_toxin HNH

QMOR01000040.1:1-12293
Bacteroidetes bacterium

? ParE_toxin ? ? Gln-synt_C Gln-synt_N ? YtxC ? SMC_N SMC_hinge SMC_N RNase_T NamZ_N NamZ_C

QMRP01000058.1:655-8737
Thermoprotei archaeon

FAD_binding_3 GGR_cat PIN ? ? DUF2283 DUF2683 ParE_toxin ParA HypA Calcineurin ?

QMXR01000120.1:255-3220
Candidatus Bathyarchaeota archaeon

? PIN MazE_antitoxin PIN VAPB_antitox RHH_3 ParE_toxin UPF0175

RFIY01000040.1:13182-26060
Candidatus Dadabacteria bacterium

? SpoVG Receiver PilZ Na_Ala_symp BATS Radical_SAM IstB_IS21 RHH_1 ParE_toxin HAD Radical_SAM HrcA_DNA-bdg HrcA GrpE Peptidase_M20 Acetylase Pribosyltran Pribosyltran_N

RDXI01000010.1:38673-51877
Bacteroidetes bacterium

GldL_N FGS PorP_SprF HEM4 DUF4271 ? SUFBD_core SufBD_N ParE_toxin ? HTH_3 Methylase_S Methylase_S Big Big Por_Secre_tail

SHNH01000008.1:41157-52305
Candidatus Saccharibacteria bacterium

? Glyco_hydro_18 fn3 5_3_exonuc_N 5_3_exonuc DNA_pol_A Hydrolase MazE_antitoxin ? Antitox_RHH ParE_toxin ? Aminotran_1_2 Methyltransf_25 NUDIX PhdYeFM_antitox PIN ?

SMYC01000093.1:4226-19990
Acidobacteria bacterium

DUF4159 DUF6483 ? Trans_reg_C Sulfatase Sulfatase Beta-lactamase Metal_resist DUF6290 ParE_toxin Cu-oxidase_2 Cu-oxidase Cu-oxidase_3 ? ? Amidohydro_1 Abhydrolase_1 STY-kinase PEGA PEGA TonB_C Peripla_BP_2 ?

STGX01000028.1:1713-15595
Glycomyces paridis

TPRs TPRs TPRs TPRs G_glu_transpept DUF2277 AlkZ-like Trypsin Pro_Al_protease NAD_binding_10 Alanine_zipper MFS_1 MFS_1 TetR_N ? ? ? SIP FAD_binding_9 PhdYeFM_antitox ParE_toxin

VBOP01000117.1:3425-3422
Nitrospirae bacterium

MazE_antitoxin PemK_toxin ParE_toxin PhdYeFM_antitox FecR

VIKD01000012.1:1-8939
Planctomycetota bacterium

Calx-beta Calx-beta ? BD-FAE ? ? HTH_3 ParE_toxin PhdYeFM_antitox ? MmeI_Mtase Peptidase_M78 ? ?

NZ_JQRQ01000010.1:3419-11963
Leptospira weilii

HTH_3 ? ? ParB_N ParA-Soj-NTPase Phage_integrase Toprim_2 DNAG_N Zn_ribbon_DnaG RelB-like ParE_toxin

NZ_BAHD01000004.1:149258-170430
Kineosphaera limosa NBRC 100340

adh_short_C2 DUF1949 UPF0029 Cys_Met_Meta_PP Viral_helicase1 UvrD-helicase PNKP_ligase Calcineurin AAA_33 PhdYeFM_antitox ParE_toxin Rce1-like DUF480 Methyltransf_25 Catalase Catalase-rel FUR DUF3418 OB_NTP_bind HA2_C HA2_N Helicase_C DEAD DUF1206 DUF1206 DUF1206 Methyltransf_12 Hen1_L PAC2

NZ_CP040005.1:493217-520180
Actinomyces sp. oral taxon 171 str. F0337

DUF7946 GmrSD_N AIRS AIRS_C DUF3073 ParE_toxin ? EamA GATase_6 GATase_7 Pribosyltran UvrD_C UvrD_C UvrD-helicase MZB Helicase_C DEAD Eco57I MmeI_Mtase NYN PhdYeFM_antitox DDE_Tnp_4 HTH_Tnp_4 Helicase_C SNF2-rel_dom

NC_016051.1:378689-389760
Thermococcus sp. AM4

ABC-3 ? ? Creatinase_N Peptidase_M24 Laminin_II ParE_toxin eRF1_1 eRF1_2 eRF1_3 HEPN NTP_transf_2 AIRS_C AIRS PP2C_2 QueC RNase_HII FeoA Methyltransf_31

NZ_JADQUA010000014.1:8758-26941
Corynebacterium diphtheriae bv. mitis

GMC_oxred_C GMC_oxred_N LGFP Amidase_2 RHH_1 ABC_tran AAA_15 ABC-ATPase ABC-ATPase ABC_tran Methylase_S BCCT NAD_binding_8 GLF ParE_toxin Peripla_BP_2 FecCD AbiEii RepL AbiEi_2 RloB

NC_018012.1:1679493-1693944
Thiocystis violascens DSM 198

HTH_3 ? YgjP-like REase SWI2_SNF2 T1R_D3-like ParE_toxin RelB-like Methylase_S HsdM_N N6_Mtase RHH_5 ? ? Arm-DNA-bind_3 Phage_integrase Phage_int_M DUF6941 ?

NC_019753.1:3309165-3329209
Crinalium epipsammum PCC 9333

Na_Ca_ex Na_Ca_ex Beta-prop_WDR3_2nd Beta-prop_WDR3_1st Beta-prop_THOC3 Na_Ca_ex Na_Ca_ex Beta-prop_THOC3 STY-kinase SUS_N SUS_EPBD GT-B_Sucrose_synth Glycos_transf_1 ParE_toxin ? TPRs GOLGA2L5 His-Kinase His-Kinase His-Kinase His-Kinase GAF Receiver His-Kinase His-Kinase Receiver Peptidase_M32 ? His-Kinase His-Kinase PAS PAS PAS PAS TPRs TPRs TPRs TPRs TPRs TPRs TPRs

NZ_CP141255.1:3986612-4007315
Cyanobacterium aponinum UTEX 3222

RAMPs HTH_3 ParE_like ? LptF_LptG ? FAD_binding_3 THF_DHG_CYH_C THF_DHG_CYH ParE_toxin Abhydrolase_1 PMT_2 UvrA_inter UvrA_DNA-bind ABC_tran ABC_tran DUF512 Radical_SAM_2 PDZ_6 Csm2_III-A Csm4_C RAMPs

NC_019779.1:3408635-3423070
Halothece sp. PCC 7418

DNA_methylase ? S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 PAF-AH_p_II Abhydrolase_1 DUF1400 DHQS_C Fe-ADH Aminotran_1_2 Methyltransf_25 ParE_toxin ? HicB_lk_antitox HicA_toxin ? T5orf172 CopG_antitoxin ?

NZ_KE332518.1:1230510-1242007
Treponema maltophilum ATCC 51939

? DUF5692 Amino_oxidase FAD_binding_6 NAD_binding_1 Fer2 DUF5692 Acyltransferase DUF6290 ParE_toxin ? ? dsrm Ribonucleas_3_3 PP-binding Ribosomal_L32p BrnT_toxin BrnA_antitoxin

NZ_KB235933.1:412158-430686
Prochlorothrix hollandica PCC 9006 = CALU 1027

? BrxC_BrxD BrxC_BrxD DDE_Tnp_ISL3 HTH_Tnp_ISL3 Zn_ribbon_ISL3 ArAE_1_C ? Pyr_redox_2 AIRS AIRS_C NTP_transferase LbH_EIF2B SMC_ScpA ABC-ATPase ? Hexapep Hexapep Hexapep NTP_transf_3 DUF3226 DEAD Helicase_C WH_Lhr BrxC_BrxD ParE_toxin

NZ_KB235948.1:3848664-3864153
Kamptonema formosum PCC 10802

GATase_7 Asn_synthase DAGK_cat YegS_C PAS PAS PAS PAS T4_CASP His-Kinase PhdYeFM_antitox ParE_toxin PhdYeFM_antitox DUF6290 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans ? YafQ_toxin Ead_Ea22 ACP_PD ? DUF433 MotA_ExbB TOP6A-Spo11_Toprim

NZ_BAAAIN010000001.1:65479-78152
Microbacterium maritypicum

Trans_reg_C Receiver ? PhdYeFM_antitox ParE_toxin VanY Ala_racemase_N Ala_racemase_C Dala_Dala_lig_N Dala_Dala_lig_C Acyl_transf_3 Glyoxalase Peripla_BP_2 ABC_tran Peptidase_S9 ? ? FecCD

NZ_AQXM01000041.1:169-4063
Acaricomes phytoseiuli DSM 14247

PhdYeFM_antitox ParE_toxin ? HTH_3 ? ? ? ? ?

NZ_CP043611.1:545405-563238
Paenibacillus antarcticus

Hol_Tox DO-GTPase2 ? GAP1-N2 GAP1-M PMSR ? ? ? ? Thioredoxin_8 SduA_C ParE_toxin RDD Coa1 RDD Coa1 HARP Phage_int_SAM_4 Phage_integrase

NZ_JAJNBZ010000027.1:5246-23883
Paenibacillus profundus

FecCD FecCD HlyD_2 Peripla_BP_2 ? ? ParE_toxin Asp_protease_2 ? HTH_18 Peripla_BP_2 N-OB tRNA-synt_2 Abhydrolase_1 Abhydrolase_1 SBP_bac_1 FlgM Beta-lactamase

NZ_KB913025.1:567638-580020
Methylovulum miyakonense HT12

HTH_Tnp_4 DDE_Tnp_4 Ogg-HhH HupE_UreJ ParE_toxin RelB-like ABC_tran ABC_tran BCA_ABC_TP_C BPD_transp_2 HEATs HEATs BPD_transp_2 Peripla_BP_5 ? Y1_Tnp ? UreD

NZ_AUZJ01000028.1:1570-11768
Treponema socranskii subsp. socranskii VPI DR56BR1116 = ATCC 35536

OEP OEP RHH_3 Dak1 ACR_tran TetR_N TetR_N Dak2 ParE_toxin ? HlyD_D23

NZ_SAYG01000018.1:177136-187369
Brachyspira aalborgi

HTH_17 Receiver CarD_TRCF_RID ? MatE MatE ? YafQ_toxin DUF6290 ParE_toxin ? YafQ_toxin ? ? ? AAA ?

NZ_JAGPYT010000008.1:33961-50811
Corynebacterium sp. CCUG 51687

FMN_red ? ? ? Mu-transpos_C_2 rve IstB_IS21 IstB_IS21 ? ? ABC_tran Anti-TRAP AAA_29 UvrA_DNA-bind ParE_toxin PhdYeFM_antitox LGFP LGFP LGFP MmeI_C MmeI_TRD MmeI_Mtase MmeI_hel MmeI_N ?

NZ_JIAG01000001.1:328022-342820
Haematomicrobium sanguinis DSM 21259

MFS_1 MFS_1 FemAB Acetyltransf_6 DoxX PhdYeFM_antitox DUF6290 ParE_toxin SAICAR_synt GARS_C GARS_A GARS_N ? UxaC Peripla_BP_3 LacI AP_endonuc_2 IclR_C HTH_IclR Sugar_tr GFO_IDH_MocA GFO_IDH_MocA_C3

NZ_BAMD01000056.1:23913-35672
Saccharicrinis fermentans DSM 9555 = JCM 21142

ParE_toxin HTH_3 ParE_toxin RelB-like HTH_3 ? ? ? RelB-like Unstab_antitox ? Phage_integrase Phage_int_SAM_1 ? ? PH_LRR1 Phage_integrase

NZ_KK211106.1:108998-9982
Desulfitibacter alkalitolerans DSM 16504

? IstB_IS21 ? ? ParE_toxin ? Fic HepT-like Polbeta ? PIN PAPS_reduct ? Mu-transpos_C_2

NZ_JAVMBV010000001.1:2595093-2615363
Planktothrix agardhii NRERC-751

RNase_H_2 Phytase-like Acetyltransf_7 Acetylase HSP33 ABC-ATPase BPD_transp_1 ? Rv2175c_wHTH Aminotran_5 ? Helicase_C AAA_16 ParE_toxin RlmN_N Radical_SAM Uma2 ?

NZ_LDYE01000004.1:44217-13137
Corynebacterium renale

? LeuA_dimer IPMS_D2 HMGL-like ParE_toxin PhdYeFM_antitox MurT_C Mur_ligase_M ? ? ? SpaA RNase_T

NZ_LRUF01000021.1:583072-594327
Pseudoalteromonas arabiensis

? AHH Polysacc_deac_1 ? ? ? ParE_toxin ? ? Zn_dep_PLPC ? IMS IMS_HHH IMS_C DUF4113 Peptidase_S24 GAF PAS His-Kinase His-Kinase Receiver ?

NZ_LUUI01000127.1:101676-108783
Methylomonas lenta

DUF3368 HTH_3 Peptidase_M78 Methylase_S ? ParE_toxin Cation_efflux MerR-DNA-bind MerR_1 RelB-like Methylase_S ? REase SWI2_SNF2

NZ_JYIJ01000008.1:687-17911
Carbonactinospora thermoautotrophica

PhdYeFM_antitox ParE_toxin Phage_int_SAM_1 Phage_integrase LPP WXG100_2 ? ? Methyltransf_25 Eco57I ResIII Helicase_C ParA-Soj-NTPase

NZ_JACKZR010000813.1:23827-40657
Nostoc sp. UCD120

DUF3531 NatA_aux_su HemY_N NUDIX UPF0150 ? ParE_toxin ? UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF CorC_HlyC CBS CBS CNNM Aldo_ket_red AMP-binding AMP-binding_C PP-binding Condensation AMP-binding AMP-binding_C PP-binding Thioesterase HicA_toxin ?

NZ_FTOP01000021.1:16432-30960
Belliella pelovolcani

? ? TolB_like ArsC Aminotran_3 Peripla_BP_6 RelB-like clamp_Gag1-like ParE_toxin RMMBL Lactamase_B ? Flavin_Reduct adh_short_C2 adh_short GMP_synt_C NAD_synthase GATase ? Strep_SA_rep

NZ_FOMJ01000005.1:56138-73192
Thiohalospira halophila DSM 15071

N6_Mtase HsdM_N AAA_16 ? Methylase_S Methylase_S EcoEI_R_C Helicase_C ResIII REase ParE_toxin Thr_dehydrat_C Thr_dehydrat_C PALP DeoRC Rib_5-P_isom_A DUF2905 Radical_SAM HlyD_D23 CbiA Peptidase_M78 HTH_3 HTH_3 SprA-related

NZ_FOLQ01000058.1:1-9856
Spirosoma endophyticum

Glycos_transf_1 Glyco_transf_4 STELLO ? DUF7683 ? ParE_toxin ? Rep3_N Rep3_C ? DUF5712 ? OMP_b-brl

NZ_PGGV01000038.1:142871-163156
Glycomyces xiaoerkulensis

? RibD_C ? DUF6247 GATase Chorismate_bind Anth_synt_I_N tRNA-synt_1 Anticodon_1 DUF5915 ParE_toxin HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB DUF4878 PDDEXK_1 UvrD_C UvrD_C UvrD-helicase DrrA1-3_C ApbA_C ABC_tran PhdYeFM_antitox ? PDDEXK_1 UvrD_C UvrD-helicase Y1_Tnp

NZ_QGDQ01000001.1:212566-232249
Quadrisphaera granulorum

Rhodanese ThiF ? MFS_1 ParE_toxin PhdYeFM_antitox AsnC_trans_reg PDDEXK_1 UvrD_C UvrD-helicase MCPsignal HAMP CHASE3 MerR_1 MerR-DNA-bind NUDIX PDDEXK_1 UvrD_C UvrD_C UvrD-helicase CW_binding_2 CW_binding_2 CW_binding_2 ? DNA_binding_1 AAA_16

NZ_CAWNWM010000004.1:343081-354680
Acaryochloris thomasi RCC1774

DUF1508 DUF1508 Sacchrp_dh_C Sacchrp_dh_NADP ? HNH ABC-ATPase DUF4276 ParE_toxin ? ARG ? RecA_C RecA UvrD_C_2 AAA_30 Sigma70_r4_2 Sigma70_r3 Sigma70_r2 ? Bac_DNA_binding

NZ_QWDC01000001.1:1502127-1518362
Mucilaginibacter conchicola

Peripla_BP_6 Methyltr_RsmB-F MacB_PCD FtsX MacB_PCD FtsX MarC MFS_1 AA_permease_2 RelB-like ParE_toxin ? H_PPase AA_permease_2 ? ? ? Peptidase_S13

NZ_CP033972.1:5949399-5962069
Gordonia insulae

Iso_dh Rv2993c-like_N FAA_hydrolase tRNA-synt_1c Anticodon_2 ? ? ParE_toxin PhdYeFM_antitox Gly_rich_2 Gp77 ? ? Cutinase Phage_holin_2 MFS_1 Abhydrolase_1

NZ_CP035494.1:892304-901312
Microbacterium protaetiae

HepT-like NTP_transf_2 HepT-like NTP_transf_2 HTH_31 PhdYeFM_antitox ParE_toxin PIN HhH-GPD Abhydrolase_1 EHN HTH_11 WYL WCX Amino_oxidase FitA-like_RHH HepT-like ? APH

NZ_VDFW01000011.1:22507-33618
Amycolatopsis alkalitolerans

FeS_assembly_P PaaA_PaaC PhdYeFM_antitox ParE_toxin Receiver ANTAR BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran Peripla_BP_6 PaaA_PaaC TetR_N TetR_C_24 PaaB BPD_transp_2 ? Zn_ribbon_PaaD

NZ_FXTP01000002.1:121000-136131
Gracilimonas mengyeensis

TonB_C VanZ tRNA-synt_2b HGTP_anticodon ProRS-C_1 ? DUF4160 DUF2442 ParE_toxin PhdYeFM_antitox Receiver GerE Carboxyl_trans Peptidase_S9 DPPIV_N MMR_HSR1_Xtn MMR_HSR1 GTP1_OBG PD40 PD40 Peptidase_S9 ? YjeF_N Carb_kinase

NZ_BKAU01000001.1:2208899-2217799
Chitinophaga cymbidii

Zn_ribbon_recom Recombinase Resolvase ? SDH_protease DUF2158 ? ParE_toxin ? ? ? ? ? ? HTH_3 DNA_methylase DNA_methylase Vsr

NZ_VOSL01000140.1:18594-38937
Lujinxingia vulgaris

? RecC_C Exonuc_V_gamma PDDEXK_1 UvrD_C UvrD_C UvrD-helicase PhdYeFM_antitox ParE_toxin ? ? HTH_3 HigB-like_toxin HTH_3 Bestrophin-like Trypan_PARP RNase_H_2 UvrD_C_2 SH3_13 AAA_19 CvfB_WH S1_2 S1_2

NZ_CP043006.1:2319410-2334231
Chitinophaga sp. XS-30

CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel PepSY_TM HmuY HmuY TonB_dep_Rec_b-barrel Plug CarboxypepD_reg ? ParE_toxin ? FAA_hydrolase rhaM Amidohydro_2 HTH_18 3keto-disac_hyd Peptidase_M15 adh_short_C2

NZ_CP046973.1:2567404-2582782
Microcystis aeruginosa FD4

Proton_antipo_M Proton_antipo_N LysR_substrate HTH_1 Uma2 Glycos_transf_2 Polysacc_synt_3 Uma2 ? ParE_toxin Sulfotransfer_3 Sulfotransfer_1 HTTM ? Glycos_transf_1 Glyco_transf_4 Glyco_trans_2_3 Glycos_transf_2 Amino_oxidase

NZ_WNKF01000004.1:173956-186725
Cyclobacterium roseum

MaoC_dehydratas ? Phage_int_SAM_1 Phage_integrase DHquinase_II Aminotran_5 MarC ? ParE_toxin ? LON_substr_bdg TPR_BSK1_C SH3_3 Cytochrome_CBB3 Pyrid_ox_like RelB-like Arm-DNA-bind_5 Phage_int_SAM_5 Phage_integrase

NZ_BMIK01000013.1:121744-138137
Parapedobacter defluvii

UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C Wzz Poly_export SLBB SLBB SLBB_2 SLBB_2 SLBB SLBB HTH_DeoR AAA_14 DUF4143 Rrf2 ParE_toxin RelB-like DUF748 Ldh_2 Peptidase_M1 HupE_UreJ_2 DUF6702 SLFN_AlbA_2 HATPase_c_4 HTH_24 23S_rRNA_IVP

NZ_DASZXR010000018.1:96803-108947
Puia sp.

HTH_11 DUF4292 Cytochrom_C DUF6786 ? HTH_3 RelB-like ParE_toxin Imm33 ? ? ? GFO_IDH_MocA GFO_IDH_MocA_C2 Glucosamine_iso Acetylase

NZ_CAWPQU010000054.1:8066-23900
Phormidium tenue FACHB-1050

Ubie_methyltran ? NadA ParE_toxin PhdYeFM_antitox ? ParE_toxin DUF2281 ? ? ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Clp_N Uma2 RNaseH-CATRA+Caspase TPRs TPRs TPRs TPRs TPRs Peptidase_M48 ?

NZ_CP071974.1:2129353-2149578
Trueperella pecoris

PhdYeFM_antitox ? Lipocalin_2 TetR_N TrkA_C TrkA_N 2-Hacid_dh_C CRISPR_Cas2 Cas_Cas1 Cas9_C Cas9_Topo Cas9_b_hairpin RuvC_III HNH Cas9_a ParE_toxin TrkH ? rve rve_3 tRNA_U5-meth_tr tRNA_U5-meth_tr HrpB_C Helicase_C DEAD

NZ_AP025343.1:1195992-1208273
Paenibacillus azoreducens

DUF899 EVE EVE ? Germane Cu_amine_oxidN1 AMIN Amidase_3 Acetylase ParE_toxin GNAT_acetyltran Methyltransf_25 HxlR KIF21A_4th Sigma70_r4 Sigma70_r2 ? Spc24 BRI3BP

NZ_CP046156.1:61318-80145
Leptolyngbya sp. BL0902

Cas_Cas1 Cas_Csm6_CARF ? ? RAMPs ? ParE_toxin RelB-like RAMPs Cmr2_N Cas10-Cmr2_palm2 WCX AAA_33 SIMPL SBP_bac_5 Cas_Cmr3 ?

NZ_CP054142.1:1452816-1469100
Treponema parvum

DUF6290 DUF2281 ? PIN ? DUF4474 ? ParE_toxin RHS AMP-binding AMP-binding_C_3 MGDG_synth Glyco_tran_28_C HigB-like_toxin AMP-binding AMP-binding_C_3 ? ThiI THUMP ThiI_fer

NZ_CAWPUQ010000159.1:30338-42889
Dendronalium phyllosphericum CENA369

His-Kinase PAS Rrf2 ? ? Pps Pps Pps Pps Pps RDD ? Peptidase_M90 ParE_toxin ? RadC UPF0758_N His-Kinase His-Kinase RsbRD_N DUF7387 HicA_toxin ? HTH_26

NZ_CP074133.1:4853081-4868238
Nocardiopsis changdeensis

Arabinose_Iso_C AraA_central AraA_N Peripla_BP_4 BPD_transp_2 ParE_toxin ? Cas_Cas2CT1978 Cas_Cas1 CRISPR_assoc Cas_Cas5d Cas_CT1975 CRISPR_Cse2 ABC_tran ABC_tran LacI Peripla_BP_3 DUF397 Aldolase_II

NZ_CDRUON020000075.1:245-14397
Oscillatoria salina

DUF2391 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PhdYeFM_antitox ParE_toxin Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pps Pilin_GH LTXXQ HisKA_3 His-Kinase Receiver GerE ? Cadherin Cadherin Pro_isomerase

NZ_JAJKFK010000005.1:185036-199329
Amycolatopsis sp. GM8

p450 p450 ? ? ? ? Phage_int_SAM_4 PaaA2 ParE_toxin HAD_2 GntR HTH_17 Sigma70_r2 Sigma70_r4_2 HTH_17 AAA_25 HTH_17 Phage_int_SAM_3 Phage_integrase Phage_integrase

NZ_CAKMSA010000013.1:5442-20385
Rothia nasimurium

DUF6290 ParE_toxin BPD_transp_1 DeoRC HTH_DeoR VanZ His_Phos_1 His_Phos_1 Abi_2 HAD_2 ? NIL ABC_tran Lipoprotein_9 FCD GntR SBP_bac_5 BPD_transp_1_N BPD_transp_1 OppC_N BPD_transp_1 ABC_tran oligo_HPY

NZ_JALPTI010000001.1:5928607-5943978
Nocardiopsis dassonvillei

3HCDH 3HCDH_N PP-binding Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N Hydrolase Methyltransf_3 DUF3418 OB_NTP_bind HA2_C HA2_N Helicase_C DEAD ? ParE_toxin ? Lactonase pAdhesive_15 BioY ABC_tran CbiQ Beta-lactamase2 Beta-lactamase2

NZ_JAMXEX010000004.1:141947-161344
Ancylothrix sp. D3o

? TPRs TPR_CcmH_CycH TPRs TPRs TPRs Guanylate_cyc CHASE2 RPE65 ? Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans PhdYeFM_antitox ParE_toxin ? DUF488 Toxin_YhaV PrlF_antitoxin PAS PAS PAS His-Kinase His-Kinase DAGK_cat YegS_C CHASE3 PAS PAS PAS PAS His-Kinase His-Kinase

NZ_BAAAOL010000009.1:274537-288584
Glycomyces algeriensis

2H tRNA-synt_1b HTH_DeoR DeoRC GalP_UDP_transf GalP_UDP_tr_C Epimerase PhdYeFM_antitox ParE_toxin ? Bmp ABC_tran ABC_tran BPD_transp_2 BPD_transp_2 dCMP_cyt_deam_1 Glycos_trans_3N Glycos_transf_3 PYNP_C GalKase_gal_bdg GHMP_kinases_N GHMP_kinases_C

NZ_JAQMSG010000107.1:677-13322
Nodularia spumigena CS-591/07A

TPRs Glyco_transf_41 Glyco_transf_41 HTH_DNAJC9 ParE_toxin ? Psb28 Glyco_trans_1_4 TylF Methyltransf_24 Glycos_transf_1 Glyco_transf_4 N_methyl Pilin_GH MoCF_biosynth ? Glyco_transf_41 Glyco_transf_41

NZ_JAQMTY010000070.1:12641-23544
Spirulina subsalsa CS-330

SpoIIE GAF_3 ? Ald_deCOase Shikimate_DH ACCA PdxJ Ycf54 ParE_toxin ? ? ParE_toxin COP23 ? DUF433 Trypsin TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs

NZ_JAQKQJ020000006.1:91758-104957
Nodosilinea sp. PGN35

Mem_trans Mem_trans HicB_lk_antitox Uma2 Aldedh DUF29 ThiC_Rad_SAM ParE_toxin ? Glyoxalase_6 Fic_N Fic ? DUF4383 SSF DUF3782 DUF29

NZ_CP112874.1:4468736-4483039
Pseudanabaena galeata CCNP1313

? His-Kinase His-Kinase PAS HAMP Uma2 Yae1_N ? PhdYeFM_antitox BCA_ABC_TP_C ABC_tran ParE_toxin ? DUF3326 MFS_1 MFS_1 Pps Pps Pps Pps Pps TerC FAD_binding_3 Glyoxalase

NZ_JAMJFP010000074.1:6644-16150
Methylobacter sp.

KleE ? DUF6792 GFA ? TMA7 ParE_toxin RPA HTH_3 KORA T2SSE TrbC VirB3 ?

NZ_CP129884.1:439764-453396
Methylomonas montana

Cas_Cas5d CRISPR_assoc MazE_antitoxin PIN HTH_37 DUF6290 ParE_toxin Cas_Cas1 Cas_Cas1 Cas_Cas2CT1978 Fer4_21 FeS tRNA_bind Anticodon_3 tRNA-synt_1g ? ? SPOR Ldt_C YkuD LysM CRISPR_assoc

NZ_JAEINN010000002.1:361527-369517
Maridesulfovibrio ferrireducens

Zn_ribbon_Elf1 DUF7835 ? ASCH ? ? ? ? ParE_toxin ? ? ? ? Lar_restr_allev ? ?

NZ_JAYFUM010000014.1:143393-155569
Arcicella rigui

PIN Arginosuc_synth Arginosuc_syn_C DUF2281 PIN Semialdhyde_dh Semialdhyde_dhC_1 ? ? ParE_toxin Aminotran_3 Peptidase_S9 DPPIV_N ? LuxE MoaE DHFR_1 Formyl_trans_N ACT

NZ_JAYGIN010000004.1:91194-103788
Arcicella sp. LKC2W

ACP_syn_III ACP_syn_III_C HigB_toxin HTH_3 Fe-S_biosyn UbiA GSHPx Arg_tRNA_synt_N tRNA-synt_1d DALR_1 ParE_toxin ? ? ? Sua5_yciO_yrdC tRNA_NucTran2_2 HD PolyA_pol_RNAbd PolyA_pol AP_endonuc_2 TAT_signal adh_short_C2